

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Formative Assessment 2

Formative Assessment (FA-2) - Answer Key

Class: Class 8

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: Written Concept Paper (20 Marks)

Section I: Multiple Choice Questions (5 x 1 = 5 marks)

Q.No	Answer	Explanation
1	c	<code>my_list = []</code> creates an empty list; <code>()</code> is tuple, <code>{}</code> is dict
2	b	List index access is $O(1)$ because lists are arrays in memory
3	c	Tuples are immutable in Python; lists, dicts, and sets are mutable
4	b	Adding a new key-value pair makes the dictionary length 3
5	c	The <code>append()</code> method adds an element to the end of a list

Section II: Short Answer Questions (5 x 2 = 10 marks)

- Differentiate between a list and a tuple. Give two use cases of each.**
 - Answer:** Lists are mutable (can be modified after creation) while tuples are immutable. Lists use `[]` brackets, tuples use `()`. Use cases for lists: dynamic collections that change (shopping cart, todo list). Use cases for tuples: fixed data (coordinates, RGB values, database records).
 - Marking:** 1 mark for differentiation, 1 mark for valid use cases.
- Write the output of the list operations.**
 - Answer:** `[20, 30, 40]` and `[50, 40, 30, 20, 10]`
 - Marking:** 1 mark for first output (slice `[1:4]`), 1 mark for second output (reverse).
- Explain the difference between `append()` and `extend()` with examples.**
 - Answer:** `append()` adds a single element to the end of a list (e.g., `list1.append([4,5])` adds the list `[4,5]` as one element). `extend()` adds each element from an iterable individually (e.g., `list1.extend([4,5])` adds 4 and 5 as separate elements).
 - Marking:** 1 mark for explaining `append` with example, 1 mark for explaining `extend` with example.
- Write a Python code snippet to iterate over all key-value pairs in a dictionary using the `items()` method.**
 - Answer:**

```
my_dict = {"name": "Alice", "age": 14, "class": "8A"}
for key, value in my_dict.items():
    print(f"{key}: {value}")
```
 - Marking:** 1 mark for correct `items()` usage, 1 mark for correct loop structure.
- What is a nested dictionary? Give one real-life example where it is useful.**
 - Answer:** A nested dictionary is a dictionary where one or more values are themselves dictionaries. Example: A student database where each student has personal details as a nested dictionary: `students = {"s1": {"name": "Alice", "marks": {"math": 90, "science": 85}}}`.

- **Marking:** 1 mark for definition, 1 mark for valid real-life example.

Section III: Long Answer Questions (5 x 1 = 5 marks)

1. Write a Python program that manages a student marks database using a dictionary of lists.

- **Answer:**

```
students = {
    "Alice": [85, 90, 78, 92, 88],
    "Bob": [70, 75, 80, 65, 72],
    "Charlie": [95, 92, 88, 90, 94]
}

# Calculate averages
averages = {}
for name, marks in students.items():
    avg = sum(marks) / len(marks)
    averages[name] = avg

# Find highest and lowest
highest = max(averages, key=averages.get)
lowest = min(averages, key=averages.get)

# Display formatted table
print(f"{'Name':<10} {'Marks':<25} {'Average':<10}")
print("-" * 45)
for name, marks in students.items():
    marks_str = ", ".join(str(m) for m in marks)
    print(f"{name:<10} {marks_str:<25} {averages[name]:<10.1f}")

print(f"\nHighest average: {highest} ({averages[highest]:.1f})")
print(f"Lowest average: {lowest} ({averages[lowest]:.1f})")
```

- **Marking:** 1 mark for dictionary structure, 1 mark for average calculation, 1 mark for finding extremes, 1 mark for formatted output, 1 mark for complete program.

Part B: Hands-on Practical Lab Task (5 marks)

List Operations Program:

- **Answer:**

```
# Create list of 10 integers
nums = []
print("Enter 10 integers:")
for i in range(10):
    nums.append(int(input(f"Number {i+1}: ")))

# Sort, reverse, find max/min
sorted_nums = sorted(nums)
reversed_nums = nums[::-1]
max_val = max(nums)
min_val = min(nums)

# Calculate sum and average without built-in functions
total = 0
count = 0
for num in nums:
    total += num
```

```
        count += 1
average = total / count

# Create tuple from sorted list
num_tuple = tuple(sorted_nums)

print(f"Original: {nums}")
print(f"Sorted: {sorted_nums}")
print(f"Reversed: {reversed_nums}")
print(f"Max: {max_val}, Min: {min_val}")
print(f"Sum: {total}, Average: {average}")
print(f"Tuple: {num_tuple}")
```

- **Marking:** 1 mark for list creation, 1 mark for list methods, 1 mark for manual sum/average, 1 mark for tuple creation, 1 mark for output.

Part C: Notebook Maintenance (5 marks)

- **Neatness and organisation of notebook (2 marks):** Check for clean, organized notebook.
- **Completion of all assigned classwork and homework (2 marks):** Verify all work is complete.
- **Proper headings, dates, and indexing (1 mark):** Ensure proper formatting throughout.